Ideation Phase

Brainstorm & Idea Prioritization

Field	Details
Team Lead	Konkimalla Bhagyasri Lasya
Team Members	Bhavitha Medakayala, Yogiswari Sanapala, Divya kauluri, Nagaram praveenya
Domain	India's Agriculture Crop Production Analysis
Tools Used	Tableau
Dataset	Historical Indian Crop Production Data, 1997–2021
Published Dashboard	https://public.tableau.com/app/profile/divya.kauluri/viz/IndiaAgricultureanalysisproject

Project Context

This report presents a comprehensive visual exploration of India's agricultural cultivation using 25 years (1997–2021) of historical crop production data. Nine Tableau visualizations — a KPI Summary panel, State-wise Production Map, Year-wise Total Production chart, Trend of Agricultural Yield chart, Cultivated Area by Crop chart, State-wise Cultivated Area Comparison chart, Leading States by Production chart, Production Distribution by Unit Type donut, and Season-wise Production pie — are combined into four role-specific dashboards and a five-point guided Story, enabling stakeholders to uncover patterns, identify areas of growth or concern, and make data-driven decisions.

By harnessing the power of Tableau, this report presents the data in a visually appealing, interactive form that lets readers explore the intricacies of India's agricultural cultivation for themselves.

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk. The Farmer's View dashboard serves Rajesh through the Cultivated Area by Crop and Season-wise Production charts.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively. The Policy Analyst's View dashboard serves Ananya through the State-wise Production Map and Trend of Agricultural Yield chart.

Scenario 3: Agribusiness Investor

Vikram is an agribusiness investor looking to invest in high-growth agricultural sectors. He needs to analyze historical crop production data to identify profitable crops, emerging trends, and regions with consistent growth. By leveraging visual analytics in Tableau, Vikram seeks to minimize investment risk and make informed decisions on where and what to invest in the agricultural market. The Investor's View dashboard serves Vikram through the Leading States by Production and Cultivated Area by Crop charts.

Step 1 — Team Gathering, Collaboration and Select the Problem Statement

The three user personas (Rajesh, Ananya, Vikram) were reviewed, and the core unmet need identified across all three was the absence of a unified, visual, interactive tool for exploring India's historical crop production data. Selected problem: "How can I empower farmers, policy analysts, and investors to make data-driven agricultural decisions using historical production data from 1997–2021?"

Step 2 — Brainstorm, Idea Listing and Grouping

- Build state-wise and district-wise production maps to show regional variation
- Create seasonal trend charts (Kharif vs Rabi vs Whole Year vs others)
- Develop year-wise and yield-trend line charts to show production and yield growth or decline over time
- Build a KPI summary panel plus crop-wise and state-wise cultivated-area comparison charts to show relative crop and state importance
- Design role-specific dashboards for the Farmer, Policy Analyst, and Investor personas
- Combine all dashboards into a Tableau Story for narrative-driven presentation
- Publish to Tableau Public for free, browser-accessible sharing

Step 3 — Idea Prioritization

Idea	Impact	Feasibility	Priority
KPI summary panel	High	High	1
Year-wise Total Production chart	High	High	2
State-wise Production Map	High	High	3
Season-wise Production chart (pie)	Medium	High	4
Cultivated Area by Crop chart	Medium	High	5
Role-specific dashboards (3)	High	Medium	6
Tableau Story (narrative)	Medium	Medium	7
Publish to Tableau Public	High	High	12